SUMULONG MEMORIAL HIGH SCHOOL
Gen. Luna St. Brgy. San Roque, Antipolo City
SY. 2022-2023

ELECTRICITY

LEARNING MODULE

NAME: ___________________________________ **YEAR AND SECTION:** ___________________

SET THE TARGET

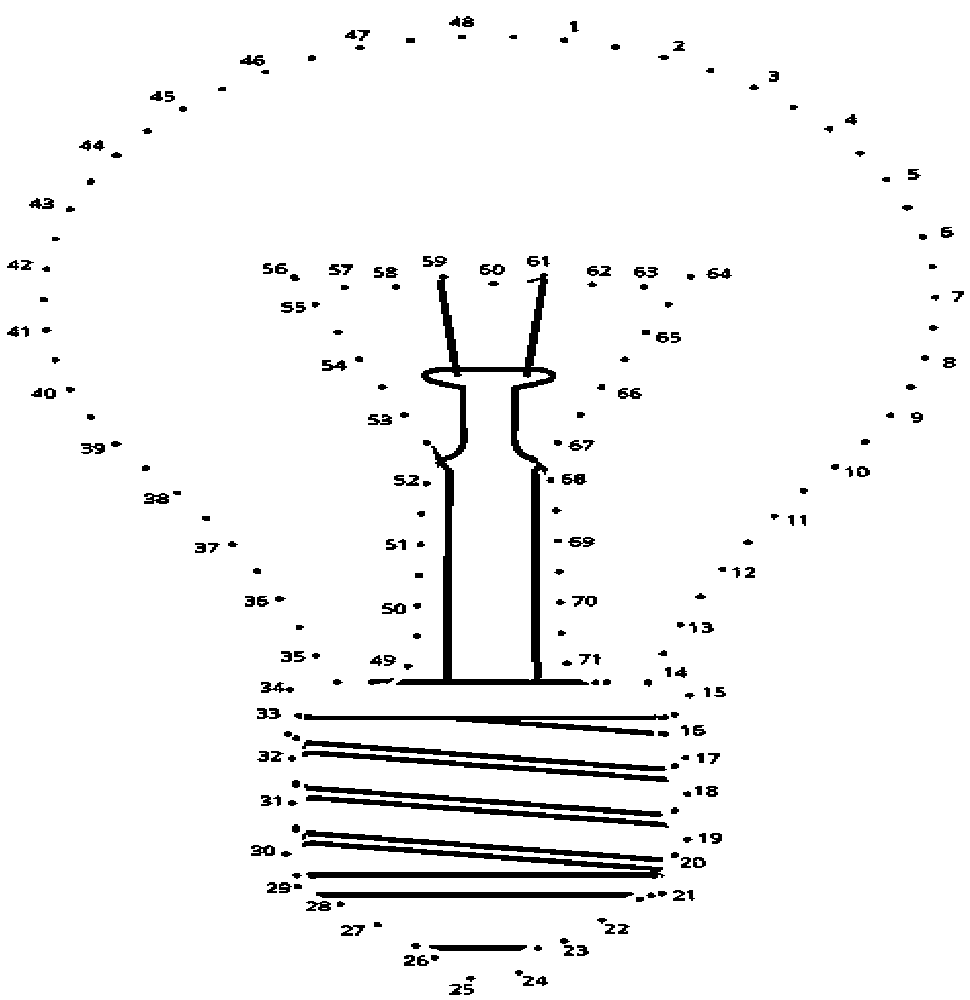
LEARNING OBJECTIVES:

At the end of this module, you are expected to:

1. Infer the relationship between current and voltage.

MOTIVATIONAL ACTIVITY

CONNECT THE DOTS! Connect the dots in order according to their numbers.



Overview:

Electricity is a part of our daily lives. Many of the activities we do every day depend on electricity. The discovery of electricity changed people's lives. It is used to heat our hot water, cook our food and keep that same food from spoiling. We use it at school and home in the form of lighting so that we may learn and enjoy leisure activities such as watching TV and playing video games. Because the use of electricity is so intertwined with our daily lives, it is important to understand the process that makes it possible.

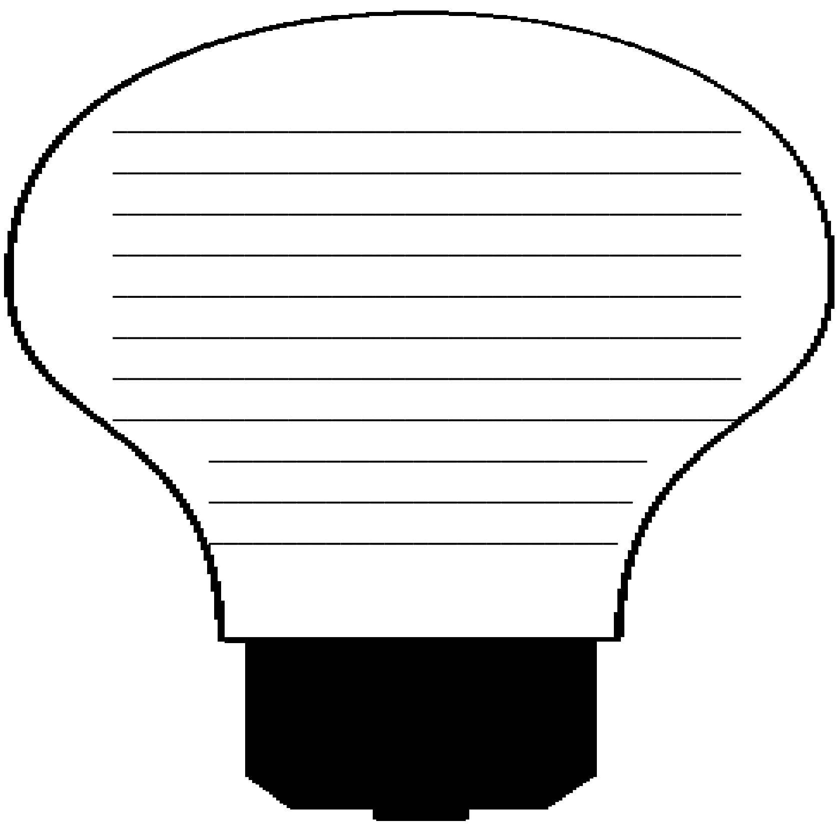
You have been learning a lot about electricity from Grade 3 to Grade 7. You have learned about its sources and uses; what materials make good conductors of electricity; what makes up an electric circuit; and how electrical energy is transferred or transformed into other forms of energy.

LEARN MORE!

Write a well-organized poem with an effective lead and ending.

In your poem, you must make a connection between the concepts of electricity and the context of your day-to-day life. You may include words such as light bulb, traffic light, street light, lamp, flashlight and even the moon, sun and stars, etc… You may write your poem in the illustration below.

"LIGHT IS EVERYWHERE"



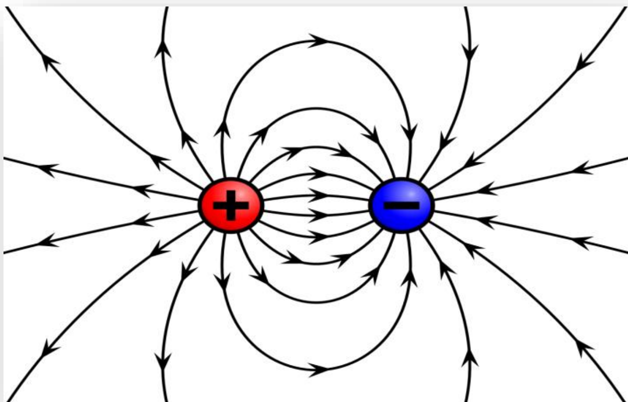
LEARN MORE!

LECTRIC CURRENT

- **ELECTRICITY** is a form of energy that can be easily changed to other forms.

THE 6 ELECTRICAL QUANTITIES

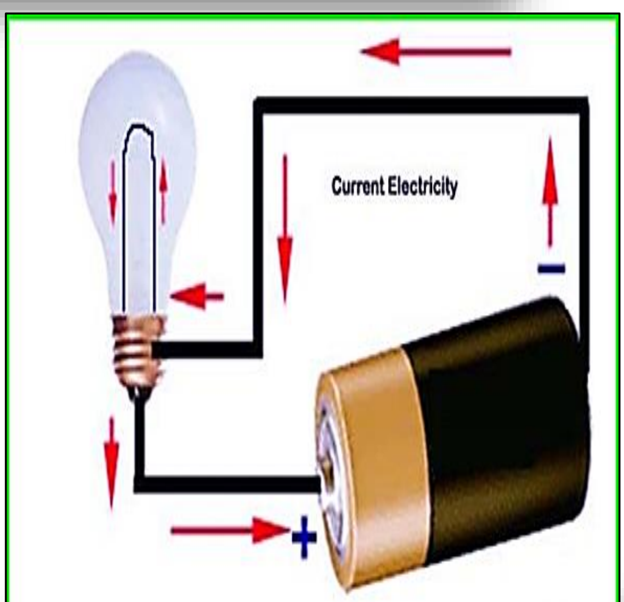
1. **ELECTRIC CHARGE** the amount of energy or electrons that pass from one body to another by different modes like conduction, induction or other specific methods. There are two types of electric charges. They are positive and negative charges.



We denote a charge y the symbol 'q' and its standard unit is **COULOMB.** Mathematically, we can say that a charge is the number of electrons multiplied by the charge on 1 electron. Symbolically, it is

$$\mathbf{Q = ne}$$

on

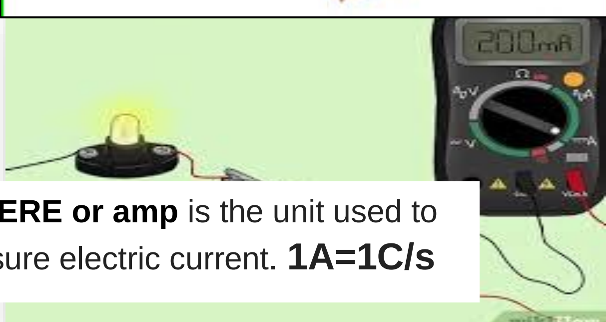


2. **ELECTRIC CURRENT** is also called the rate of flow of charge, is caused by a stream of electrons moving from the negative voltage terminal toward the positive terminal. Approximately 1020 electrons pass through a wire in each second.

When an amount of charge ΔQ passes through a surface (like the cross-section of a wire) in a time Δt, the current I is:

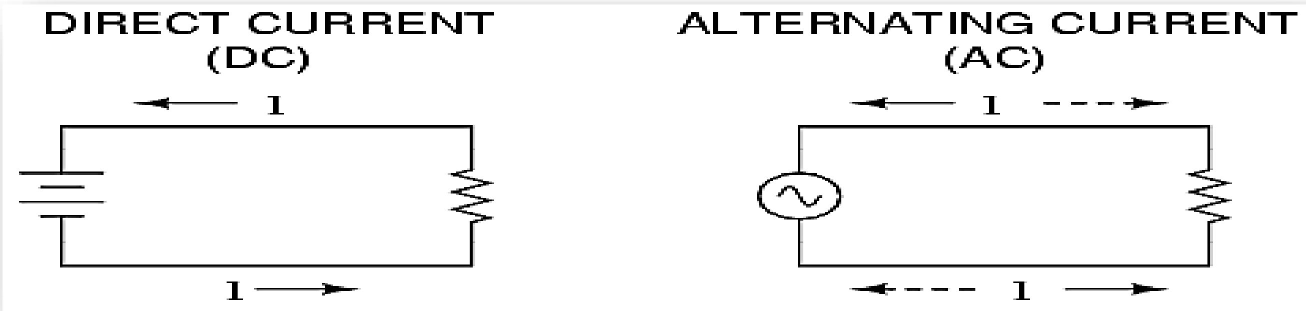
$$\mathbf{I = \frac{\Delta Q}{\Delta t}}$$

The unit, ampere (A), is named after Andre-Marie Ampere, a French physicist who made important contributions to the theory of electricity and magnetism.



AMPERE or amp is the unit used to measure electric current. **1A=1C/s**

•TWO FORMS OF ELECTRIC CURRENT:

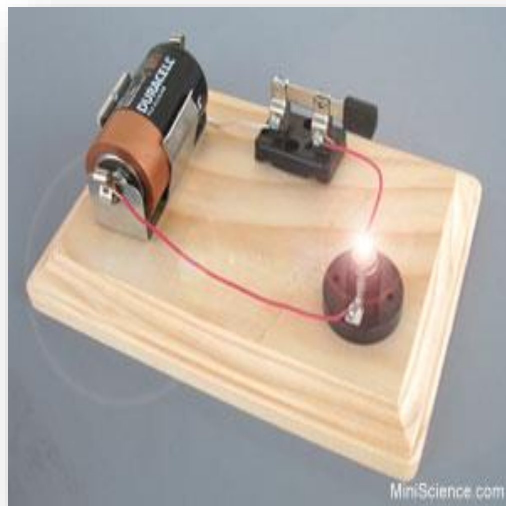


DIRECT CURRENT
It is generated, the electron flow in one direction only, which is from negative to the positive terminal of the source.

ALTERNATING CURRENT
It produced by constantly changing voltage from positive to negative to positive and so on.

3. ELECTROMOTIVE FORCE (emf)

BATTERY is a device which converts chemical energy into electrical energy. It consists of a cell or several cells connected in series or parallel.The
emf of the battery is the total voltage developed by the chemical reaction, but not all of this total voltage is available for doing useful work in an eternal circuit.
TERMINAL VOLTAGE voltage which is supplied to the external circuit, that is,
Terminal voltage = emf – internal voltage drop



- When a cell supplies current to an external circuit, it will be found that the voltmeter no longer reads open-circuit voltage (emf) of the cell. The reason is that part of the emf is used in forcing current through the resistance of the cell and the remainder is used in forcing current through the external

VOLTMETER is an instrument used for measuring electric potential difference between two points in an electric circuit.**VOLT** is named after the Italian physicist **ALLESANDRO VOLTA** (1745-1827) who did research with some of the original batteries.

VOLTAGE is a measure of the energy available to produce a flow of charges through a circuit. **(1V=1J/C)**

$$V\frac{W}{Q}$$

Mathematically:

circuit. Expressed as an equation:

$$V_{emf} = V_1 + lr$$ **(Equation 1)**

$$V_{emf} = lR + l$$ **(Equation 2)**

$$V_{emf} = l(R + r)$$

V = emf of the cell or group of cell
V_1**=** voltage measured across the terminal
I = current
r = resistance

4. **RESISTANCE** is a measure of the opposition to current flow in an electrical circuit. **Resistance** is measured in ohms, symbolized by the Greek letter omega **(Ω)**. **OHMS** are named after **GEORG SIMON OHM** (1784-1854), a German physicist who studied the relationship between voltage, current and **resistance**.

$$\frac{R_1}{R_2} = \frac{L_1}{L_2}$$ **(EQUATION 1)**

$$\frac{R_1}{R_2} = \frac{A_2}{A_1}$$ **(EQUATION 2)**

$$R_1 = R_0\,(1 + \alpha\Delta T)$$ **(EQUATION 2)**

R_1 R_2 **=** are the resistance of conductors.

L_1 L_2 = length

A_2 A_1 = cross sectional areas

R_0 = resistance at 0°C

T = temperature

OHMMETER is an electrical instrument that measures electrical resistance (the opposition offered by a substance to the flow of electric current).

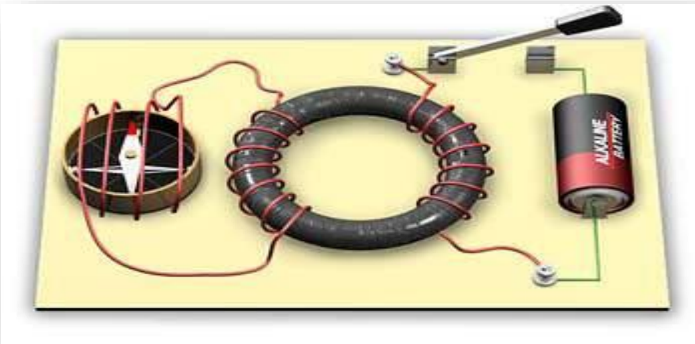
The **Resistor Color Code** Calculator decodes and identifies a value and tolerance of 4 band wire wound resistors.

This **color code** indicates the value, rating and tolerance. **Resistors** have standard **colors** for identification of the resistance.

The Resistor Colour Code Table

Colour	Digit	Multiplier	Tolerance
	0	1	
Brown	1	10	± 1%
Red	2	100	± 2%
Orange	3	1,000	
Yellow	4	10,000	
Green	5	100,000	± 0.5%
Blue	6	1,000,000	± 0.25%
Violet	7	10,000,000	± 0.1%
Grey	8		± 0.05%
White	9		
Gold		0.1	± 5%
Silver		0.01	± 10%
None			± 20%

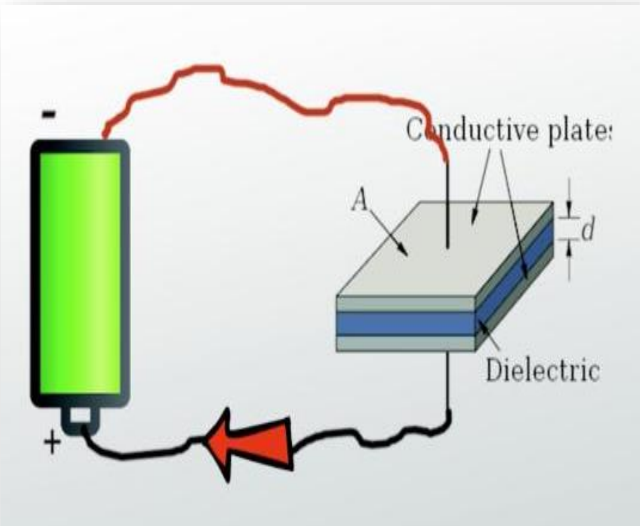
5. INDUCTANCE



Inductance is the ability of an inductor to store energy and it does this in the magnetic field that is created by the flow of electrical current.

L=VI t

6. CAPACITANCE



Capacitance is the ability of a component or circuit to collect. The **capacitance** of the capacitor tells you how much charge it can store when connected to a particular battery. and store energy in the form of an electrical charge.

If electric charge is transferred between two initially uncharged conductors, both become equally charged, one positively, the other negatively, and a potential difference is established between them. The capacitance *C* is the ratio of the amount of charge *q* on either conductor to the potential difference *V* between the conductors, or simply

C = Capacitance in farads
Q = charge in Coulombs
V = voltage of the battery in volts

ELECTRICAL QUANTITIES, FORMULA AND UNITS

QUANTITY	FORMULA	UNIT
Electric Charge, q	Q=ne	Coulomb
Electric Current, I	$I = \frac{\Delta Q}{\Delta t}$	Ampere
Electromotive force	$V\frac{W}{Q}$	Volt
Resistance, R	$R = \rho\frac{L}{A}R_1 = R_0\,(1 + \alpha\Delta T)$	Ohm
Inductance, L	L=VI t	Henry
Capacitance, C	$C = \frac{Q}{V}$	Farad

NAME: ________________________________ **YEAR AND SECTION:** __________________

ACTIVITY 1: BUILD A FLASHLIGHT

OBJECTIVE:

At the end of the activity, you will be able to

- Use the concepts of electric circuits and build a working flashlight

MATERIALS:

One cardboard tube, flashlight bulb, paper cup, scissors, one D cell, aluminum foil, duct tape 10cm wire

PROCEDURES:

1. Insert d cell inside the cardboard tube to check if the fit is right. After removing the battery from the tube, punch two holes on the side (near the middle) of the tube about 3 cm apart.
2. Connect a 10-cm wire to each terminal of the battery using small strips of duct tape. Touch the loose ends of the wires to a flashlight bulb to identify where to connect them.
3. Wrap an aluminum foil on a paper cup. Poke a hole in the bottom of the paper cup that is slightly smaller than the bulb.
4. Secure the base of the light bulb through the hole.
5. Pass the long wire through one of the holes in the tube. Tape it to the side of the tube, leaving 2-cm outside the tube. The other end should reach the end of the tube.
6. Insert the battery in the tube. Be sure to pass the wire attached to the bottom of the battery through the other hole of the tube. Check if the two wires outside the tube can touch.
7. Secure the battery inside the tube using the duct tape.
8. Attach the wires from the end of the tube to the contact the points on the bulb. Make sure that all connections are tight.
9. Connect the two free ends of the wires together. Did the bulb light? If it doesn't check to make sure that all connection.

QUESTIONS FOR ANALYSIS:

1. Compare your flashlight to commercial one. Is there any difference in their function? Explain.
2. Design a more convenient witch for your flashlight. Have your teacher approve your switch design. How will you build and test your switch?

CONCLUSION

From this activity, what can you conclude about force and its relation to the object's motion?

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Concept Micro

Electricity is a form of energy that can be easily changed to other forms.
Electrical quantities include the following: (1) Electric Charge,q; (2)Electric Current,I; (3) Electromotive force or potential difference, V; (4)Resistance, R; (5) Inductance, L; and (6) Capacitance, C.

TEST YOURSELF!

TEST I. **MULTIPLE CHOICE.** Write the letter of the best answer on the blank before each number.

_________1. Which device called cell that converts chemical energy to electric energy?
a. Switch
b. Paper
c. Battery
d. Plastic

_________2. It is the rate of charge caused by a stream of electrons moving from negative toward positive terminal.
a. Electric current c. Electromotive force
b. Electric charge d. Capacitance

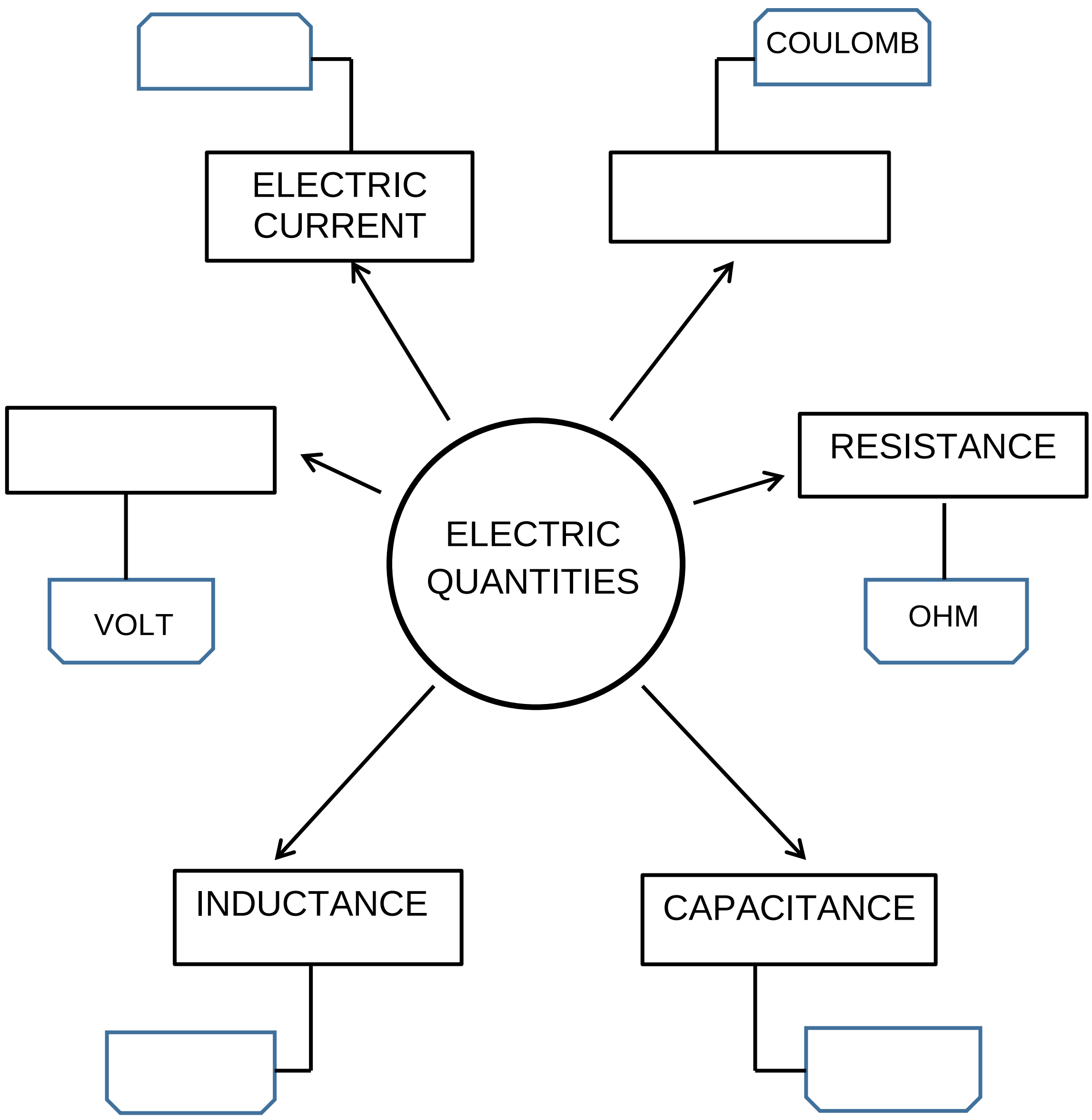
_________3. Which of the following is the unit of Inductance?
a. Farad c. Coulomb
b. Henry d. Ohm

_________4. What is the function of the resistor color code?
a. Indicate the rating, value and tolerance of the resistor
b. Calculate the electrical charge
c. Converts energy of resistor
d. Indicates the power of voltage

_________5. What are the two forms of electric current?
a. AB & CD c. OC & PC
b. DC & AC d. QC & AC

TEST YOURSELF!

TEST II. CONCEPT MAP Complete the concept map below by filling the missing word



GLOSSARY OF TERMS

CAPACITANCE is the ability of a component or circuit to collect. The **capacitance** of the capacitor tells you how much charge it can store when connected to a particular battery and store energy in the form of an electrical charge.

ELECTRIC CHARGE the amount of energy or electrons that pass from one body to another by different modes like conduction, induction or other specific methods. **COULOMB** is the standard unit.

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ELECTRICITY is a form of energy that can be easily changed to other forms.

ELECTROMOTIVE FORCE (emf) of the battery is the total voltage developed by the chemical reaction, but not all of this total voltage is available for doing useful work in an eternal circuit. **VOLT** is the unit used.

INDUCTANCE is the ability of an inductor to store energy and it does this in the magnetic field that is created by the flow of electrical current. **HENRY** is the standard unit.

RESISTANCE is a measure of the opposition to current flow in an electrical circuit. **OHM** is the standard unit.

References:

Printed Books

Madriaga, Estrelita A. et. al.,2015. Worktext in Science and Technology Science Links. Sampaloc Manila: REX Book Store Publishing House.